Statement of Qualifications
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Describe your experience managing and administering server environments.

My experience comes from my Homelab. I installed Windows Server 2022 using VMware and have it sitting behind a virtualized pfSense firewall. The firewall made it possible to create L3 segmentation between my real home network and the lab network. The goal with Windows Server was to configure DNS/DHCP services and get a feel for how it ties into AD. Using Linux and Windows VMs I was able to successfully join them to the AD domain. I experimented with creating/deleting user accounts, resetting passwords, and enforcing password complexity policies. I created different OUs and practiced moving users between them as well as editing properties to simulate updating user information. Using Ubuntu Server on my main network at home, I configured self-hosted services such as cloud storage (Nextcloud), a media server (Jellyfin), and an SMB fileshare. Connecting to those services outside of my local network is a breeze with Tailscale.

Provide an example of a complex issue you diagnosed and resolved.

One of my courses at WGU involved diagnosing and troubleshooting simulated tickets from real-world scenarios, and one of those tickets involved solving connectivity issues for subnets. Certain subnets were able to reach the internet and internal resources, and some were not. After reviewing the EXOS switch configurations and utilizing traceroute, I was able to pin-point where traffic was being held up. Turns out there were 2 routers with incorrectly configured routing protocols. I backed up the current configuration, wiped it, and reconfigured it to operate on OSPF since this was a small single area network. The proper routes were picked up in the neighbor discovery process and connectivity was restored. This issue could have also been solved by manually adding static routes, but with OSPF properly configured, new devices can be added to the network and have the routing tables updated dynamically.

Explain a time where you implemented improvement, automation, or process enhancement.

Multiple user accounts needed to be created on EXOS switches, Windows 10 PCs, and Ubuntu Desktop PCs. Creation of these accounts can be done manually, but that’s incredibly tedious, especially if creating accounts becomes a regular duty. I created Python scripts to automate the creation of multiple VLANs on the EXOS switches, and an Ansible playbook to create each account across the 3 different OS environments. Since Ansible operates under the concept of idempotence, it's safe to addon more users to the script and run it repeatedly when necessary. While not technically automated, I did create an update/upgrade script for the Ubuntu users, so if updates were run manually in the terminal, I could just execute a simple script instead of typing out the full commands.